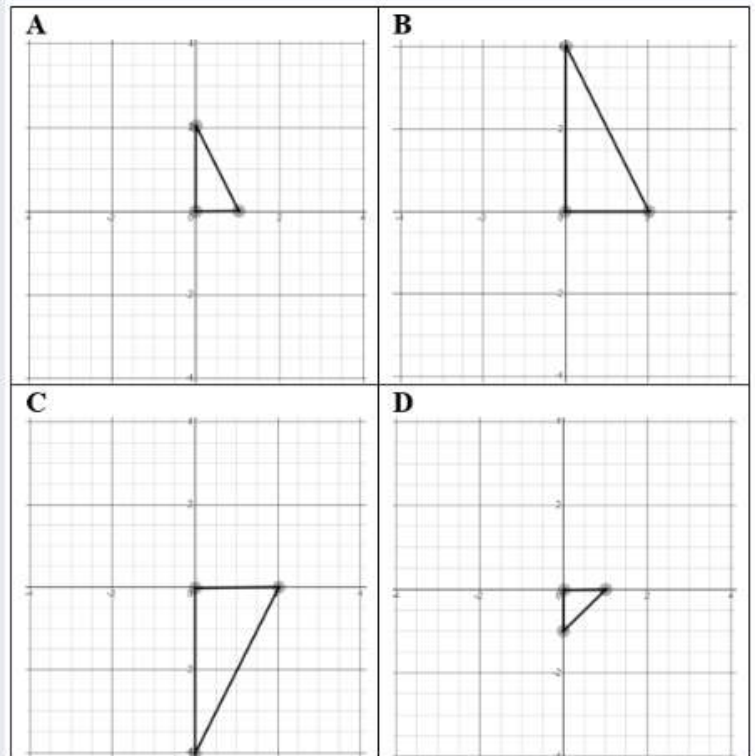


8

Assume that drawTriangle function draws a triangle (0,0), (1,0), (0,2), what is the output of executing the following code segment?

```
glMatrixMode(GL_PROJECTION);  
glLoadIdentity();  
glOrtho(-10.0, 10.0, -10.0, 10.0, -2.0, 2.0);  
glMatrixMode(GL_MODELVIEW);  
glPushMatrix();  
glLoadIdentity();  
glScalef(-1.0, 1.0, 1.0);  
glRotate(180, 0, 0, 1);  
glScalef(2.0, 2.0, 1.0);  
drawTriangle();  
glPopMatrix();
```



A

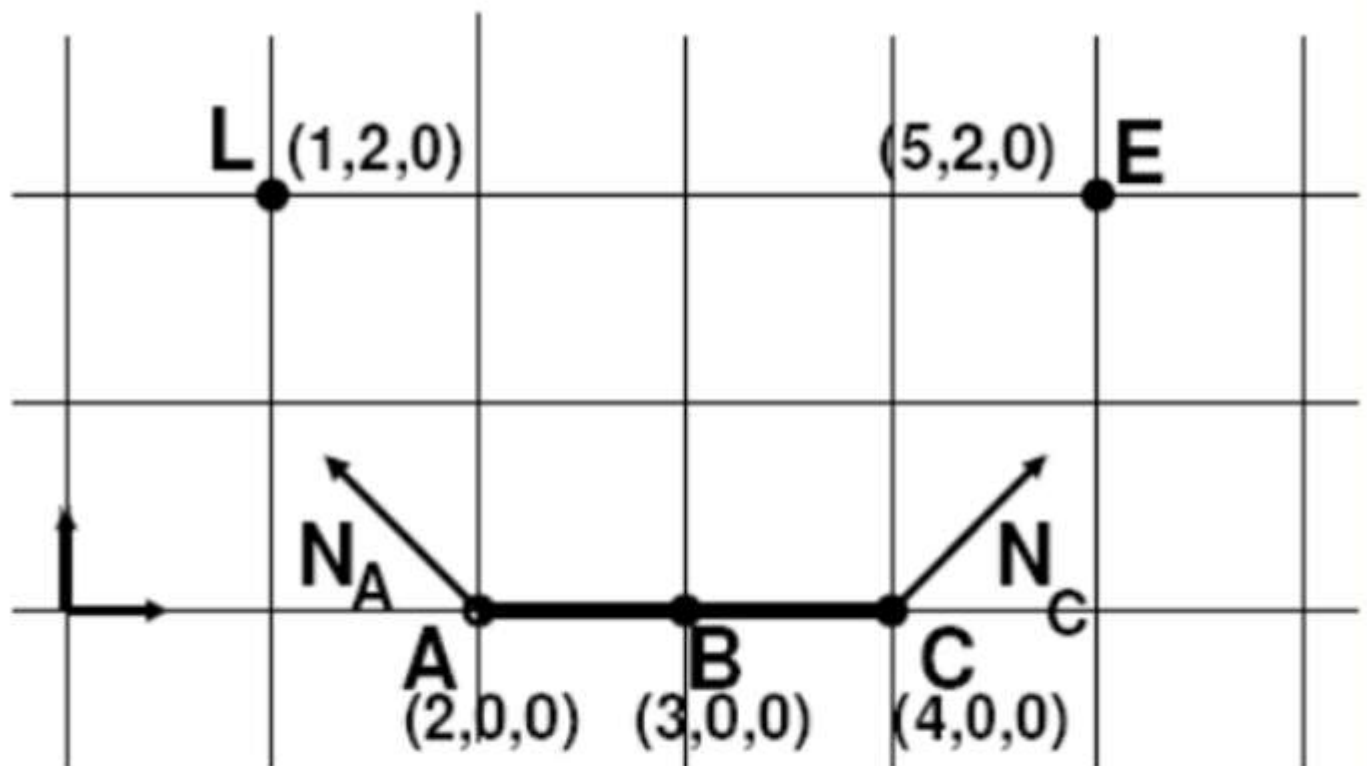
B

C

D

In the following image, the light source and the eye are denoted by L and E respectively. The light source emits diffuse light = $(1.0/3.0, 0.5/3.0, 1.0/3.0)$, spot cutoff = 180, and spot exponent = 2.0. On surface AC the normal at A and C are $N_A = (-1, 1, 0)$ and $N_C = (1, 1, 0)$ respectively. Everything is drawn to scale. Assume the constant attenuation = 0, linear attenuation = 0, quadratic attenuation = 1, the material ambient = $(0, 0, 0)$, the material emission = $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for all points. Using the Phong reflection model, answer the following question:

The normalized normal vector $N_C =$



☒ $((-1)/\sqrt{2}, 1/\sqrt{2}, 0)$

☐ $(1/\sqrt{2}, (-1)/\sqrt{2}, 0)$

☐ $(1/\sqrt{2}, 1/\sqrt{2}, 0)$

☐ $(1, 1, 0)$

10

With color tracking, you can tell OpenGL to set material properties by only calling glColor. 

☒ True

☐ False

11

The position of the vertex after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(1.0f, 1.0f, 0.0f);  
glRotatef(90.0, 0, 0, 1.0);  
glScalef (-2.0,1.0, 1.0);  
glBegin(GL_POINTS);  
    glVertex3f(1.0, 0.0, 0.0);  
glEnd(); 
```

☒ (1,0,0)

☐ (-1,0,0)

☐ (0,1,0)

☐ (1,-1,0)

☐ (-1,-1,0)



Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glScalef(-1.0, 1.0, -1.0);  
glScalef(-2.0, -1.0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glTranslatef(-1.0f, -2.0f, -3.0f);
glRotatef(90.0, 0, 0, 1.0);
glPushMatrix();
glTranslatef(-1.0f, -2.0f, -3.0f);
glRotatef(-90.0, 0, 0, 1.0);
glScalef (-1.0,0.5, -1.0);
glPopMatrix();
glRotatef(90.0, 0, 0, 1.0);
glTranslatef(-1.0f, -2.0f, 3.0f);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 2 \\ -1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

14

The command that takes a texture and creates a hierarchy of coarser and coarser versions of it is 

☒ gluBuild2DMipmaps

☐ glTexCoord2f

☐ glGenTextures

☐ glBindTexture

☐ glTexParameteri

15

In modern OpenGL, the user may supply shaders which will execute on the GPU. 

☒ True

☐ False



16

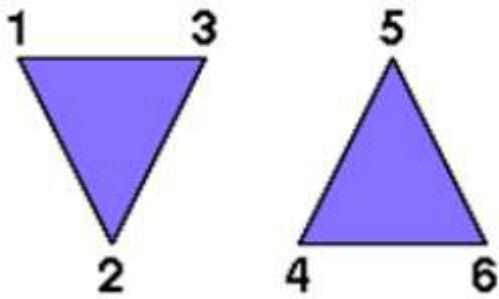
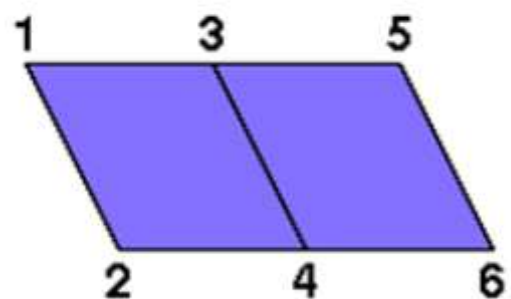
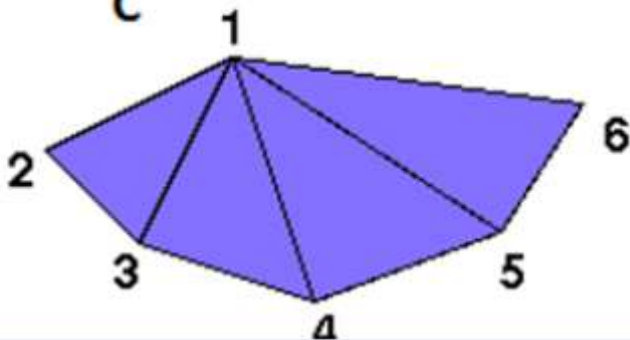
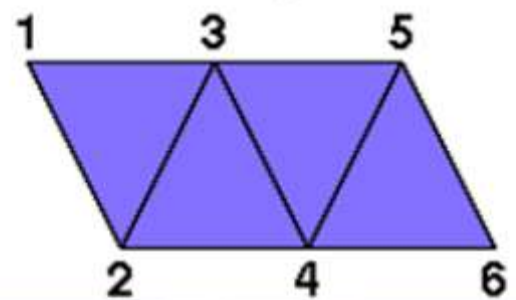
The cross product of two vectors that are orthogonal is the zero vector. 

☒ True

☐ False

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_TRIANGLE_STRIP);  
  glVertex2fv(v1); glVertex2fv(v2);  
  glVertex2fv(v3); glVertex2fv(v4);  
  glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```

A**B****C****D**

A

B

C

D

18

In the following code segment using OpenGL Shading Language (GLSL), what is the value d?

```
vec4 TestVec = vec4(1.0, 2.0, 3.0, 4.0);  
vec2 d = TestVec.zx; 
```

d= NULL

d=(3.0, 1.0)

d=(1.0, 2.0, 3.0, 4.0)

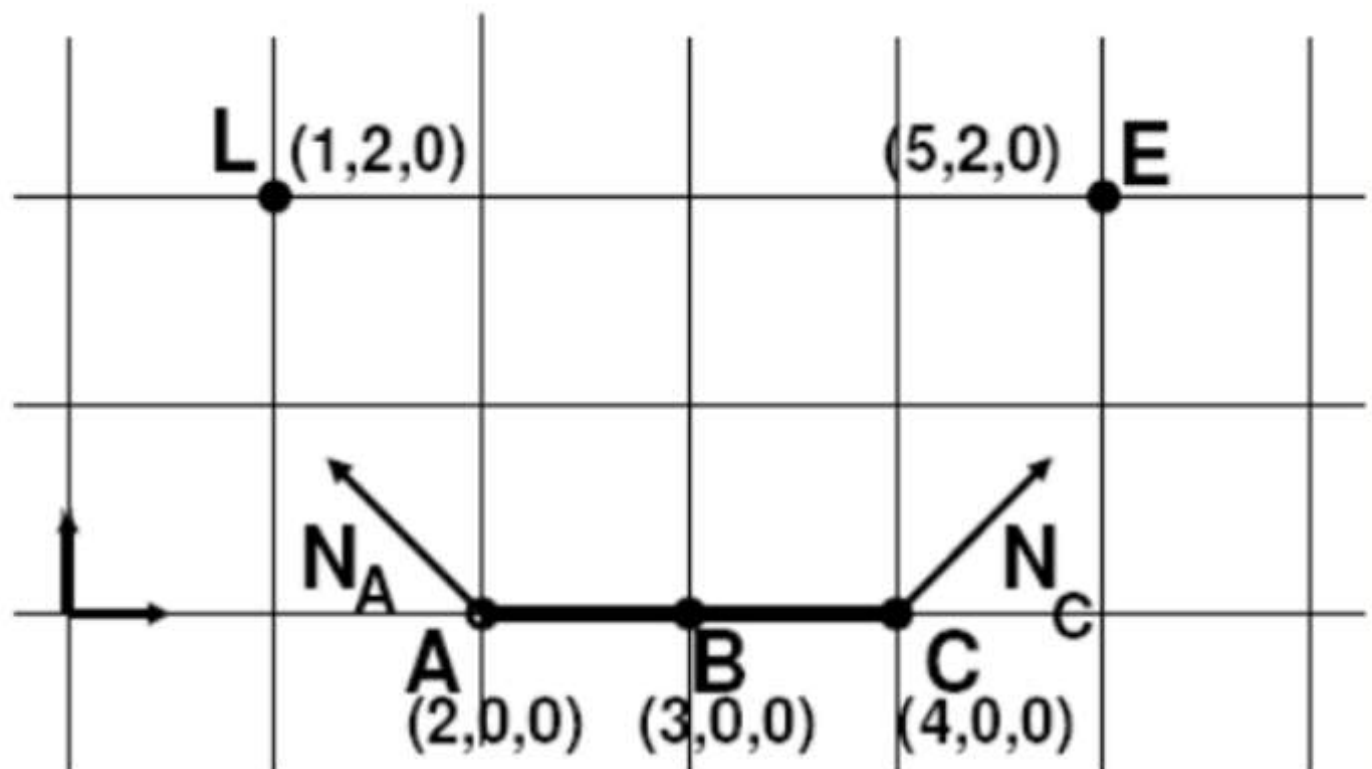
d=(1.0, 3.0)

d=(4.0, 2.0)

syntax error

In the following image, the light source and the eye are denoted by L and E respectively. The light source emits diffuse light = $(1.0/3.0, 0.5/3.0, 1.0/3.0)$, spot cutoff = 180, and spot exponent = 2.0. On surface AC the normal at A and C are $N_A = (-1, 1, 0)$ and $N_C = (1, 1, 0)$ respectively. Everything is drawn to scale. Assume the constant attenuation = 0, linear attenuation = 0, quadratic attenuation = 1, the material ambient = $(0, 0, 0)$, the material emission = $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for all points. Using the Phong reflection model, answer the following question:

The diffuse intensity for the point A =



$$1/(2\sqrt{10})$$

$$3/\sqrt{5}$$

$$2/\sqrt{5}$$

$$3/\sqrt{10}$$

20

OpenGL can draw convex polygons. 

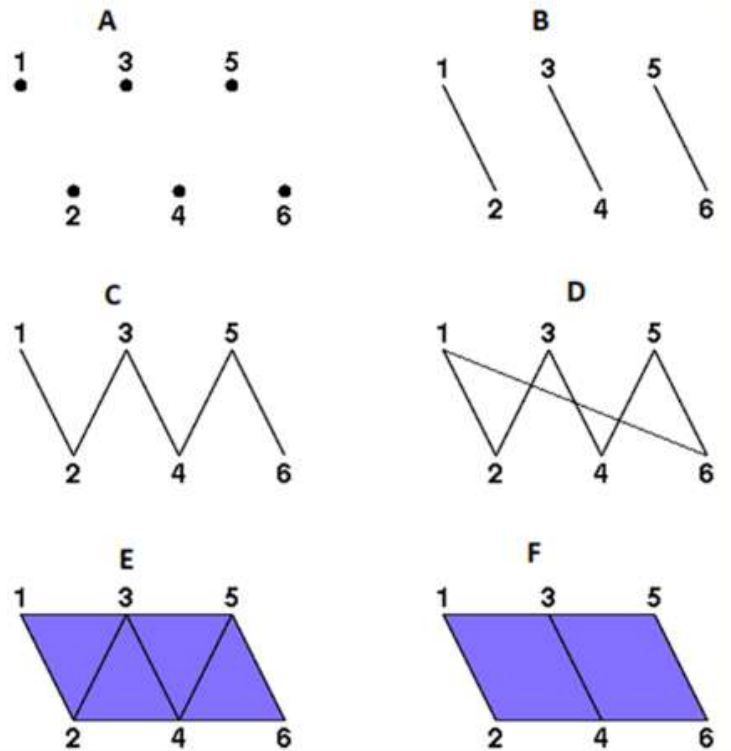
 True

False

21

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_LOOP);  
  glVertex2fv(v1); glVertex2fv(v2);  
  glVertex2fv(v3); glVertex2fv(v4);  
  glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```



A

B

C

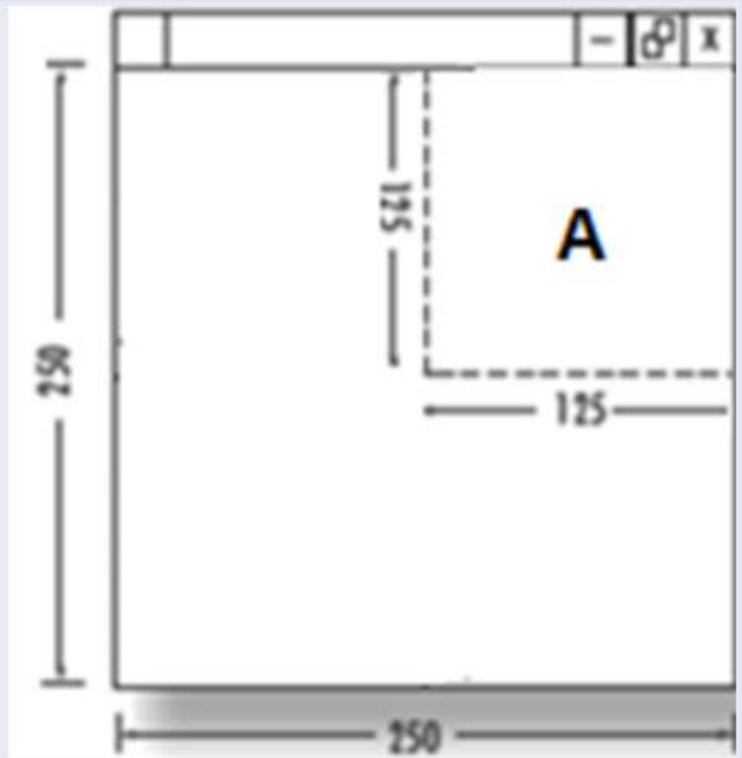
D

E

F

22

To map the output of the OpenGL code to area A (as shown in the figure), we use the `glViewport`



```
glViewport(0, 0, 125, 125);
```

```
glViewport(125, 125, 125, 125);
```

```
glViewport(125, 125, 0, 0);
```

```
glViewport(0, 125, 0, 125);
```

23

The position of the vertex after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(-1.0f, -1.0f, 0.0f);  
glRotatef(90.0, 0, 0, 1.0);  
glBegin(GL_POINTS);  
    glVertex3f(1.0, 0.0, 0.0);  
glEnd();
```

(1,0,0)

(-1,0,0)

(0,1,0)

(1,-1,0)

(-1,-1,0)

24

OpenGL version 4.x does not support shaders. 

☒ True

☐ False

25

The OpenGL function used to set up a parallel view volume is 

☒ glViewport()

☐ gluPerspective()

☐ glDisplayfunc()

☐ glOrtho()



26

How do you activate a shader program in OpenGL? 

☒ glUseProgram(ProgramID);

☐ glProgram(ProgramID);

☐ glShader(ProgramID);

☐ glUseShader(ProgramID);

27

In OpenGL Shading Language, attribute parameters are passed from vertex shader to fragment shader. 

☒ True

☐ False

27

In OpenGL Shading Language, attribute parameters are passed from vertex shader to fragment shader. 

☒ True

☐ False

28

What is a varying variable? 

☒ A variable passed from the C code to the vertex shader

☐ A variable passed from the C code to the fragment shader

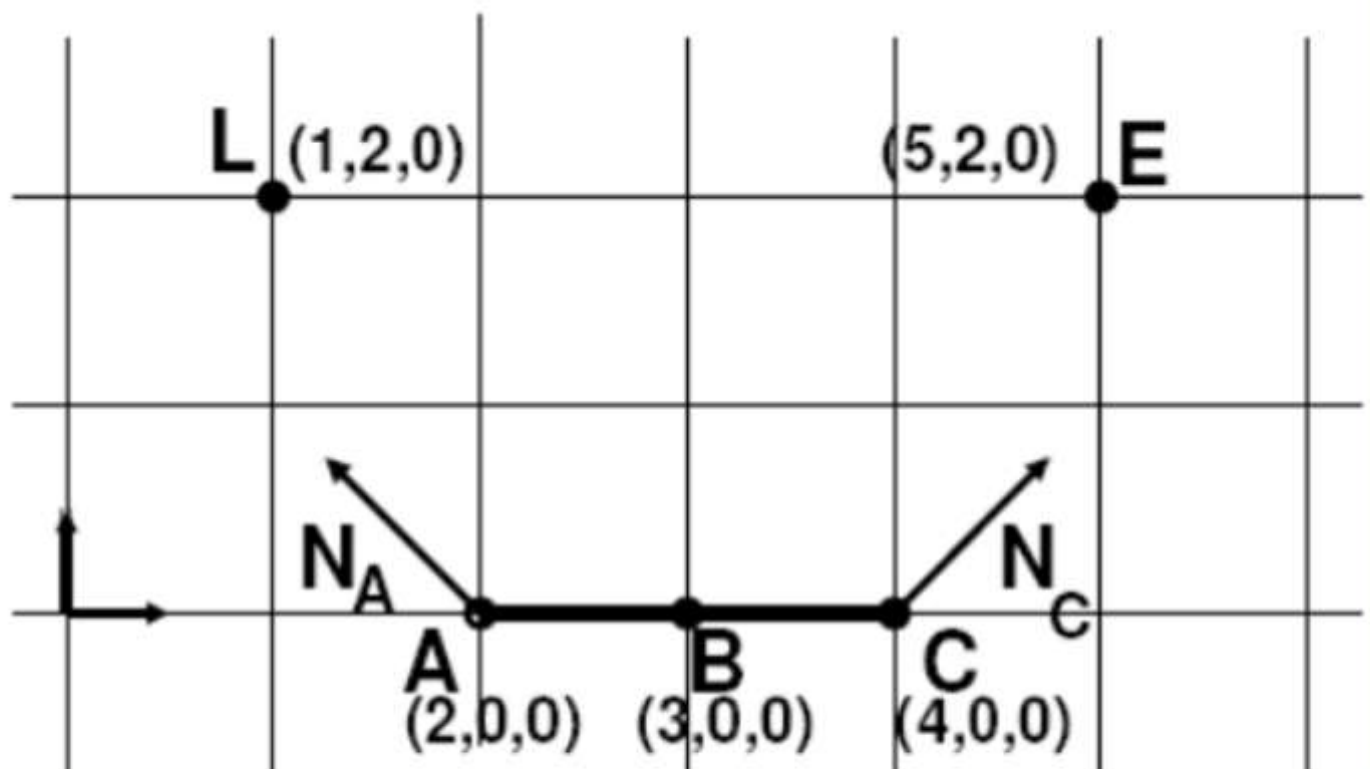
☐ A variable passed from the vertex shader to the fragment shader

☐ A variable passed from the fragment shader to the vertex shader

☐ None of the above

In the following image, the light source and the eye are denoted by L and E respectively. The light source emits diffuse light = $(1.0/3.0, 0.5/3.0, 1.0/3.0)$, spot cutoff = 180, and spot exponent = 2.0. On surface AC the normal at A and C are $N_A = (-1, 1, 0)$ and $N_C = (1, 1, 0)$ respectively. Everything is drawn to scale. Assume the constant attenuation = 0, linear attenuation = 0, quadratic attenuation = 1, the material ambient = $(0, 0, 0)$, the material emission = $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for all points. Using the Phong reflection model, answer the following question:

The attenuation factor for the point C =



5

 $1/5$ $1/13$

13

30

What type of vertex shader variable has the same value for each vertex? 

- uniform variable
- varying variable
- lighting variable
- attribute variable

31

Order the following shader processes: 

- 1 **glCreateProgram**
- 2 **glAttachShader**
- 3 **glLinkProgram**
- 4 **glUseProgram**

32

When using the Phong Reflectance Model, we calculate the red, green, and blue color channels independently. 

☒ True

☐ False

33

Specular reflections are view independent. 

☒ True

☐ False

34

What components make up a light source in OpenGL? 

Specular and Ambient

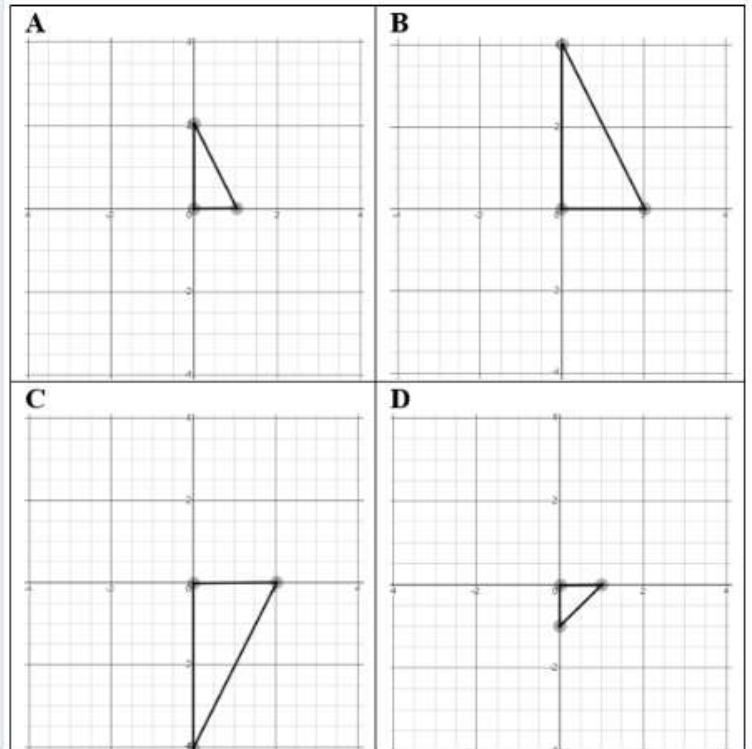
Diffuse, Specular, and Ambient

Diffuse and Ambient

Diffuse, Opaque, and Ambient

Assume that drawTriangle function draws a triangle (0,0), (1,0), (0,2), what is the output of executing the following code segment?

```
glMatrixMode (GL_PROJECTION);
glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0);
glMatrixMode(GL_MODELVIEW);
glPushMatrix();
glLoadIdentity();
glScalef(-1.0, -1.0, 1.0);
glRotate(180, 0, 0, 1);
glScalef(2.0, 2.0, 1.0);
drawTriangle();
glPopMatrix();
```



A

B

C

D

36

OpenGL runs only on Microsoft operating systems. 

☒ True

☐ False

37

Texture mapping with mipmapping consumes more memory than texture mapping without mipmapping. 

☒ True

☐ False

38

How many triangles are drawn if we use the drawing mode `GL_TRIANGLE_STRIP` with seven vertices between `glBegin(GL_TRIANGLE_STRIP)` and `glEnd()`? 

☒ 2

☐ 3

☐ 4

☐ 5

39

Backface culling refers to eliminating geometry with backfacing normals. 

☒ True

☐ False

40

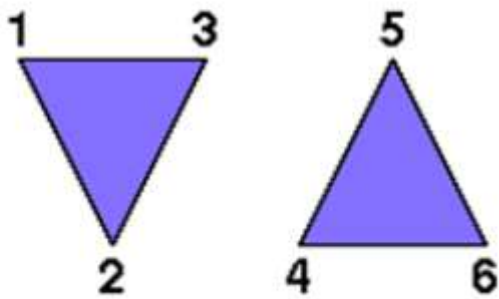
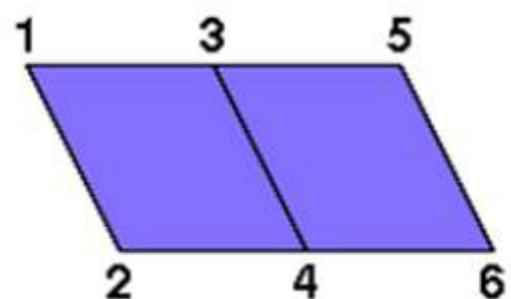
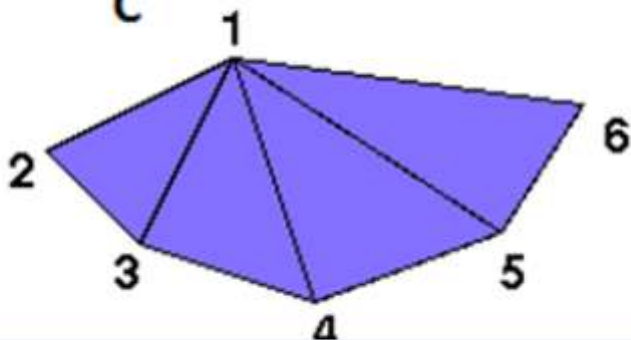
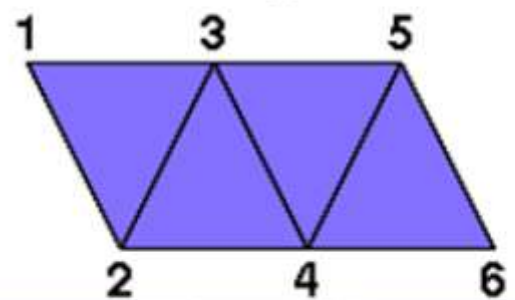
One important rule when you use quads is that all four corners of the quadrilateral must lie in a plane. 

☒ True

☐ False

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_QUAD_STRIP);  
  glVertex2fv(v1); glVertex2fv(v2);  
  glVertex2fv(v3); glVertex2fv(v4);  
  glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```

A**B****C****D**

A

B

C

D

42

Which values can be used with `glTexParameter` and `GL_TEXTURE_MAG_FILTER` parameter names? 

`GL_FAR`

`GL_NEAREST`

`GL_BILINER`

`GL_MIPMAP`

43

What is the purpose of `GL_REPEAT`? 

To have a texture repeat in a direction (horizontal or vertical)

To repeat the last command

To enable double-buffering

To restart the application

To have a command repeat until stopped.



44

The RGB of a yellow color is 

☒ (1.0,1.0,0.0)

☐ (1.0,0.0,1.0)

☐ (0.0,1.0,1.0)

☐ (1.0,1.0,1.0)

45

The positional light source is appropriate for a distant light source, e.g., the sun. 

☒ True

☐ False

46

Rotating the 3D point $(-1,0,0)$ by 90 around z-axis gives what: 

☒ (1,0,0)

☐ (-1,0,0)

☐ (0,1,0)

☐ (0,-1,0)

☐ (-1,-1,0)

47

Which of the following is not a valid argument to `glMatrixMode`? 

☒ GL_PROJECTION

☐ GL_DEPTH

☐ GL_MODELVIEW



48

What are the three major types of shaders available in OpenGL 3.X 

☒ Filtered, non-filtered, and Exponent.

☐ Vertex, Fragment, and Geometry

☐ Vertex, Fragment, and Isomorphic

☐ Vertex, Fragment, and Anisotropic

49

All rotation matrices are invertible. 

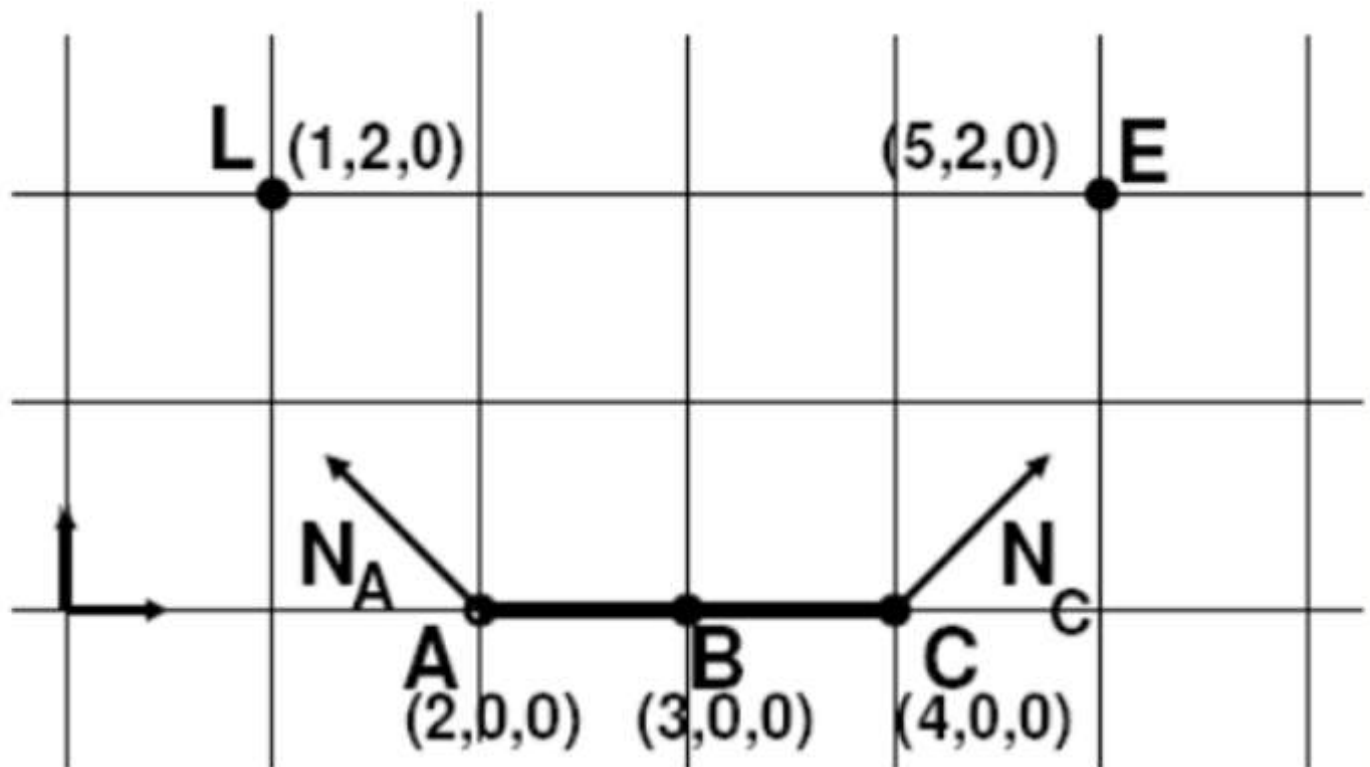
☒ True

☐ False



In the following image, the light source and the eye are denoted by L and E respectively. The light source emits diffuse light = $(1.0/3.0, 0.5/3.0, 1.0/3.0)$, spot cutoff = 180 , and spot exponent = 2.0 . On surface AC the normal at A and C are $N_A = (-1, 1, 0)$ and $N_C = (1, 1, 0)$ respectively. Everything is drawn to scale. Assume the constant attenuation = 0 , linear attenuation = 0 , quadratic attenuation = 1 , the material ambient = $(0, 0, 0)$, the material emission = $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for all points. Using the Phong reflection model, answer the following question:

The spotlight effect for the point A =



0

 $1/2$

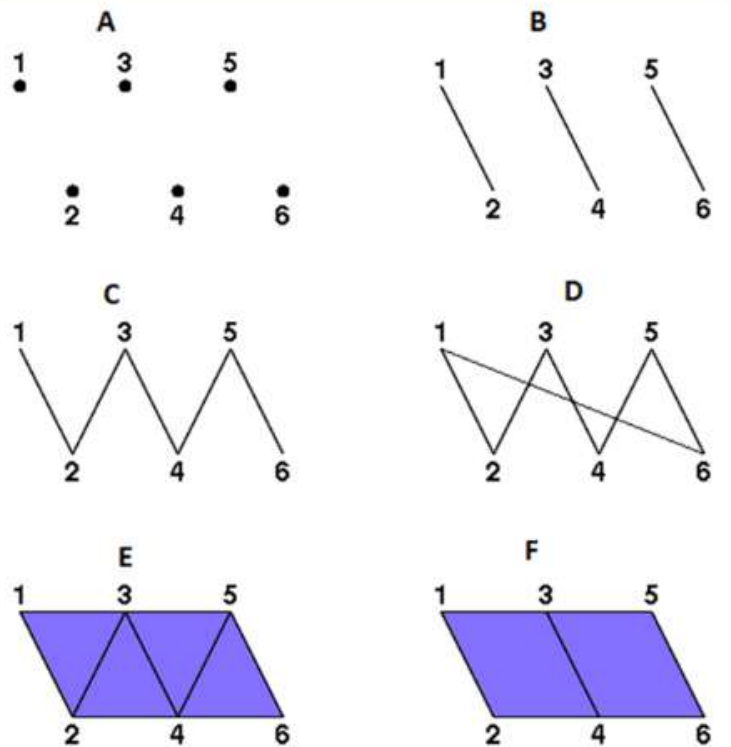
1

2

51

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_STRIP);  
  glVertex2fv(v1); glVertex2fv(v2);  
  glVertex2fv(v3); glVertex2fv(v4);  
  glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```



A

B

C

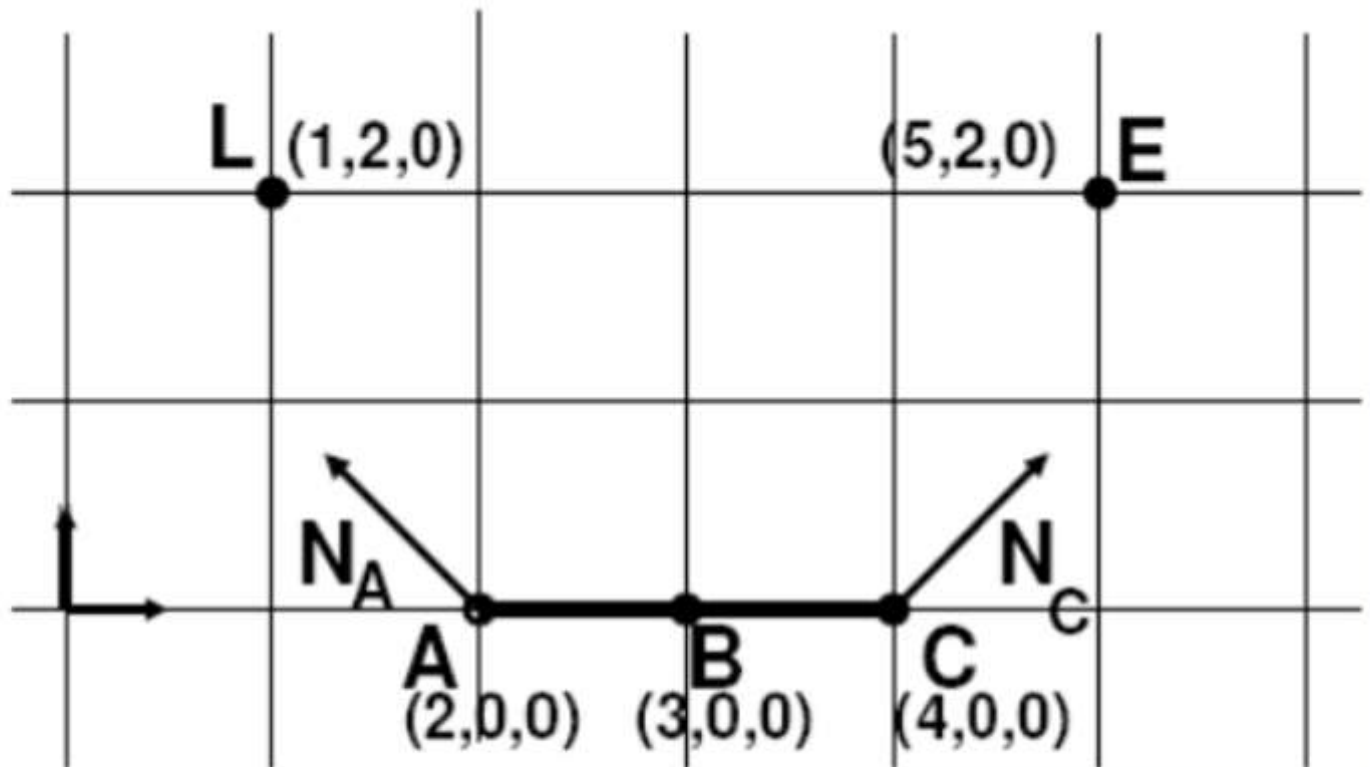
D

E

F

In the following image, the light source and the eye are denoted by L and E respectively. The light source emits diffuse light = $(1.0/3.0, 0.5/3.0, 1.0/3.0)$, spot cutoff = 180, and spot exponent = 2.0. On surface AC the normal at A and C are $N_A = (-1, 1, 0)$ and $N_C = (1, 1, 0)$ respectively. Everything is drawn to scale. Assume the constant attenuation = 0, linear attenuation = 0, quadratic attenuation = 1, the material ambient = $(0, 0, 0)$, the material emission = $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for all points. Using the Phong reflection model, answer the following question:

The colour of diffuse component for the point A =



$$(1/(2\sqrt{10}), 1/(2\sqrt{10}), 1/(2\sqrt{10}))$$

$$(1, 1, 1)$$

$$(1/(2\sqrt{5}), 1/(2\sqrt{5}), 1/(2\sqrt{5}))$$

$$(1/\sqrt{10}, 1/\sqrt{10}, 1/\sqrt{10})$$

53

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(30.0, 0, 0, 1.0);
glRotatef(60.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

54

OpenGL supports z-buffering.

True

False

55

If you had 1,000,000 vertices and wanted to change all the vertex positions by multiplying each x-coordinate by the same vertex's y-coordinate, the fastest place to do it is: ☐

☒ with a c++ function

☐ in the vertex shader

☐ in the fragment shader

56

OpenGL Shading Language (GLSL) supports pointers. ☐

☒ True

☐ False

57

Is the following code, correct?

```
glStart(GL_TRIANGLES);  
glVertex2f(0.0f, 0.0f);  
glVertex2f(25.0f, 25.0f);  
glVertex2f(50.0f, 0.0f);  
glEnd(); ☐
```

☒ Yes

☐ No

10

With color tracking, you can tell OpenGL to set material properties by only calling glColor. 

☒ True 

☐ False

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glScalef(-1.0, 1.0, -1.0);  
glScalef(-2.0, -1.0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☒ B

☐ C

☐ D

15

In modern OpenGL, the user may supply shaders which will execute on the GPU. 

☒ True

☐ False



16

The cross product of two vectors that are orthogonal is the zero vector. 

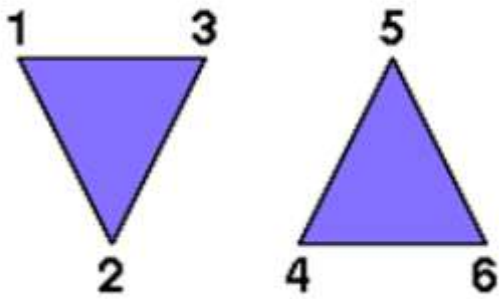
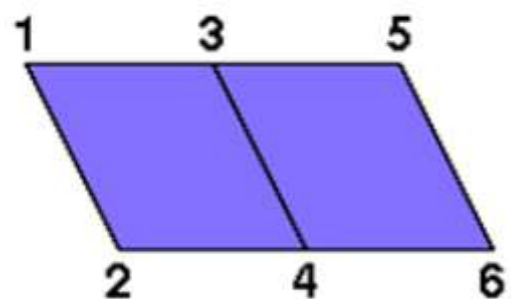
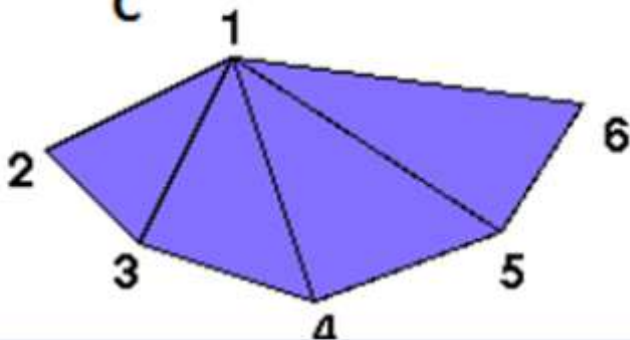
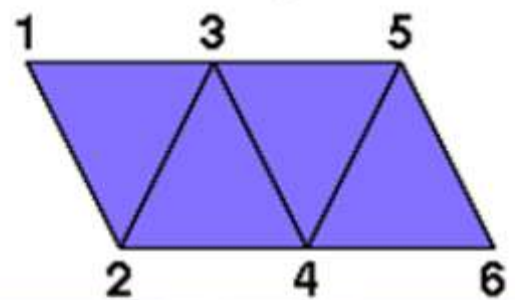
☐ True

☒ False



Select the output of executing the following OpenGL code segment:

```
glBegin(GL_TRIANGLE_STRIP);  
  glVertex2fv(v1); glVertex2fv(v2);  
  glVertex2fv(v3); glVertex2fv(v4);  
  glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```

A**B****C****D**☐ A☐ B☐ C☒ D

20

OpenGL can draw convex polygons. 

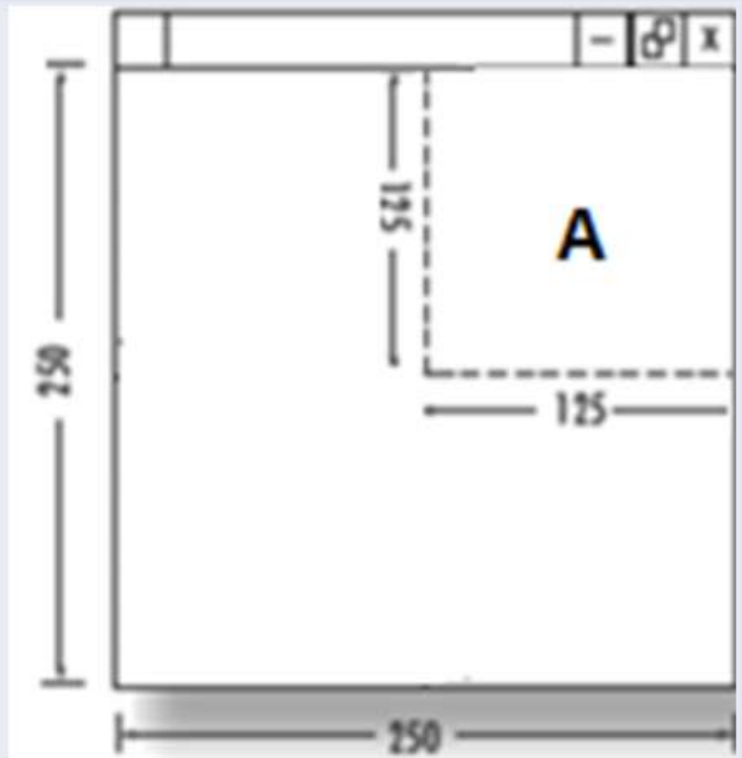
True



False

22

To map the output of the OpenGL code to area A (as shown in the figure), we use the `glViewport`



```
glViewport(0, 0, 125, 125);
```

```
glViewport(125, 125, 125, 125);
```

```
glViewport(125, 125, 0, 0);
```

```
glViewport(0, 125, 0, 125);
```

24

OpenGL version 4.x does not support shaders. 

☐ True

☒ False



25

The OpenGL function used to set up a parallel view volume is 

☐ glViewport()

☐ gluPerspective()

☐ glDisplayfunc()

☒ glOrtho()



What is a varying variable? 

- ☐ A variable passed from the C code to the vertex shader
 - ☐ A variable passed from the C code to the fragment shader
 - ☒ A variable passed from the vertex shader to the fragment shader
 - ☐ A variable passed from the fragment shader to the vertex shader
 - ☐ None of the above
- 

30

What type of vertex shader variable has the same value for each vertex? 

uniform variable 

varying variable

lighting variable

attribute variable

31

Order the following shader processes: 

1 **glCreateProgram**

2 **glAttachShader**

3 **glLinkProgram**

4 **glUseProgram**



32

When using the Phong Reflectance Model, we calculate the red, green, and blue color channels independently. 

☒ True

☐ False

34

What components make up a light source in OpenGL? 

- ☐ Specular and Ambient
- ☒ Diffuse, Specular, and Ambient
- ☐ Diffuse and Ambient
- ☐ Diffuse, Opaque, and Ambient

36

OpenGL runs only on Microsoft operating systems. 

☐ True

☒ False



37

Texture mapping with mipmapping consumes more memory than texture mapping without mipmapping. 

☒ True

☐ False



38

How many triangles are drawn if we use the drawing mode `GL_TRIANGLE_STRIP` with seven vertices between `glBegin(GL_TRIANGLE_STRIP)` and `glEnd()`? 

☒ 2

☐ 3

☐ 4

☒ 5

39

Backface culling refers to eliminating geometry with backfacing normals. 

☐ True

☒ False

40

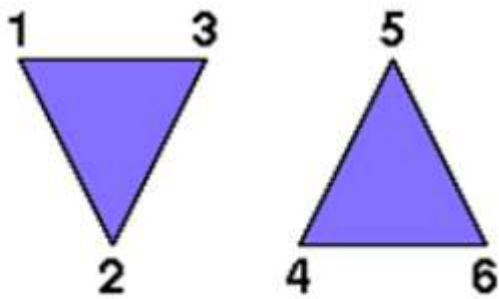
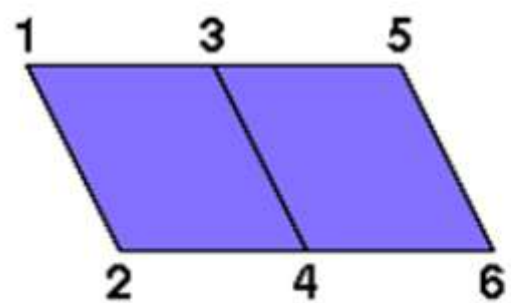
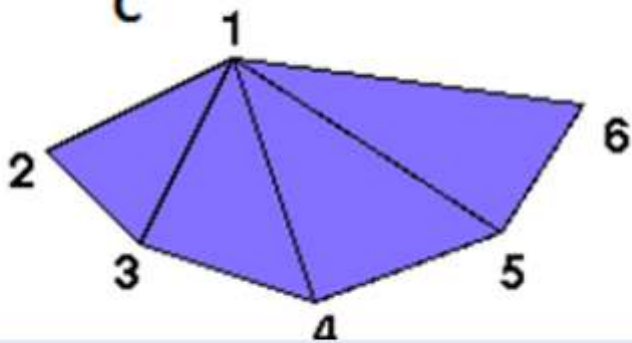
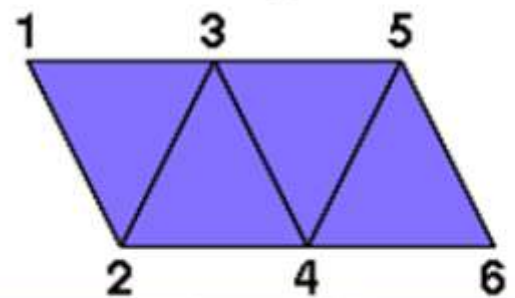
One important rule when you use quads is that all four corners of the quadrilateral must lie in a plane. 

☒ True

☐ False

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_QUAD_STRIP);  
  glVertex2fv(v1); glVertex2fv(v2);  
  glVertex2fv(v3); glVertex2fv(v4);  
  glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```

A**B****C****D**☐ A☒ B☐ C☐ D

43

What is the purpose of GL_REPEAT? 

- ☒ To have a texture repeat in a direction (horizontal or vertical)
- ☐ To repeat the last command
- ☐ To enable double-buffering
- ☐ To restart the application
- ☐ To have a command repeat until stopped.



44

The RGB of a yellow color is 

☒ (1.0,1.0,0.0)

☐ (1.0,0.0,1.0)

☐ (0.0,1.0,1.0)

☐ (1.0,1.0,1.0)

46

Rotating the 3D point $(-1,0,0)$ by 90 around z-axis gives what: 

☐ (1,0,0)

☐ (-1,0,0)

☐ (0,1,0)

☒ (0,-1,0) ✓

☐ (-1,-1,0)

47

Which of the following is not a valid argument to `glMatrixMode`? 

☐ GL_PROJECTION

☒ GL_DEPTH ✓

☐ GL_MODELVIEW



48

What are the three major types of shaders available in OpenGL 3.X 

Filtered, non-filtered, and Exponent.

Vertex, Fragment, and Geometry 

Vertex, Fragment, and Isomorphic

Vertex, Fragment, and Anisotropic

49

All rotation matrices are invertible. 

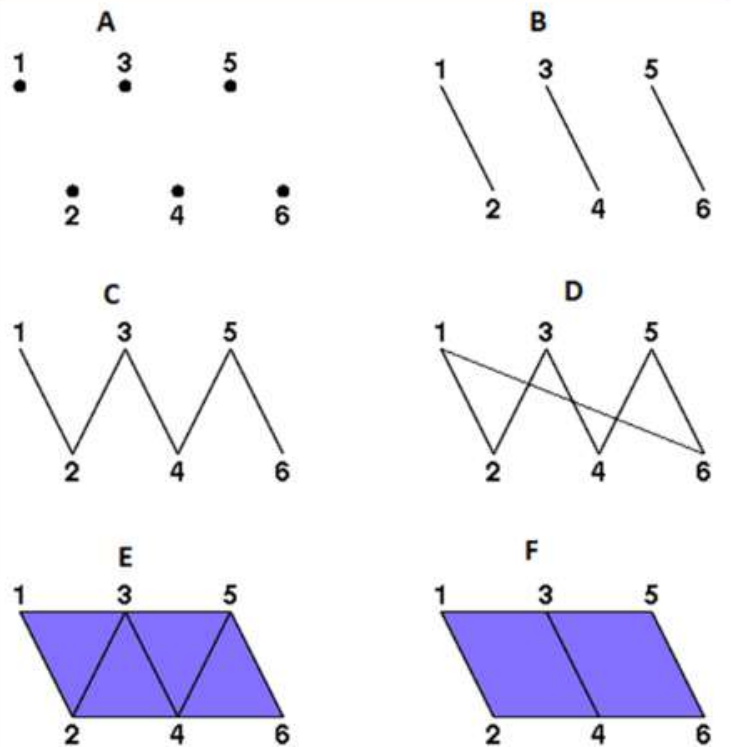
True 

False

51

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_STRIP);  
  glVertex2fv(v1); glVertex2fv(v2);  
  glVertex2fv(v3); glVertex2fv(v4);  
  glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```



- ☐ A
- ☐ B
- ☒ C
- ☐ D
- ☐ E
- ☐ F

53

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(30.0, 0, 0, 1.0);
glRotatef(60.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

54

OpenGL supports z-buffering.

True

False

56

OpenGL Shading Language (GLSL) supports pointers. 

True

False 

57

Is the following code, correct?

```
glStart(GL_TRIANGLES);  
glVertex2f(0.0f, 0.0f);  
glVertex2f(25.0f, 25.0f);  
glVertex2f(50.0f, 0.0f);  
glEnd(); 
```

Yes

No 